Title: Application of remote sensing and GIS in identifying and assessing the spatial distribution of microplastics

Thesis statement

Micro plastic (MP) has now been recognized as an emerging pollutant widely spread in the water, land, and air. Studies have shown that there are significant negative health impacts of exposing to microplastics in humans as well as in other living beings. However, research up to now have been more concerned about aquatic and land-based MPs while limited research exists on the existence and fate of airborne MPs [1].

Abbasi (2021) stated that the MPs in the atmosphere could be a possible source of MPs in the oceans and the land [2]. Even though the airborne microplastics being a significant potential source for MPs in water and land, a few research have yet been conducted on the occurrence and fate of micro plastics in the atmosphere. Therefore, it is a timely need conducting research in this field.

The need of standardized methods in sampling and measuring MPs in the atmosphere ensuring the reproducibility and comparability of results has been stressed in some studies [3]. The need of understanding the spatial and temporal variation of MP worldwide has also been emphasized. There are a few attempts made in using hyperspectral imagery for detecting varieties of polymers and they are still in the development process [4]. Hence, finding new methodologies in assessing the type, quantity, and the behaviour of MPs in the atmosphere is a matter of great concern.

Therefore, this study focuses on assessing the potential of using the remote sensing technology and GIS in identifying and assessing the spatial distribution of airborne microplastics.

Research questions

1. What kind of available remote sensing data can be used in detection of microplastics in the atmosphere?
2. How can remote sensing data be combined with field measurements of microplastic types, their sizes and abundance in the atmosphere?
3. How can remote sensing data be used in finding microplastic transport pathways in the atmosphere through spatial scales?

References

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